

First Aid and Medical Emergencies — Comprehensive Guide

1. Principles of First Aid

First aid is the immediate assistance given to any person suffering a sudden illness or injury, with care provided to preserve life, prevent the condition from worsening, and promote recovery, until professional medical help arrives.

The primary survey (DRABC): Danger (ensure scene is safe), Response (check consciousness), Airway (open and maintain), Breathing (check for breathing), Circulation (check pulse, control bleeding).

Call for help early — 112 (India emergency number). Do not delay in calling. Activate EMS immediately for: cardiac arrest, severe breathing difficulty, uncontrolled bleeding, suspected spinal injury, unconsciousness, chest pain, stroke symptoms.

Universal precautions: use gloves if available. If not, use clothing, bags, or other barriers when handling blood or body fluids. Wash hands thoroughly before and after providing care.

2. Cardiac Arrest and CPR

Recognising cardiac arrest: person is unresponsive, not breathing normally (absent or gasping breaths — agonal breathing is NOT normal breathing).

CPR technique (adult): place heel of hand on the centre of the chest (lower half of sternum). Place other hand on top, interlace fingers. Compress straight down 5-6 cm at a rate of 100-120 compressions per minute. Allow full chest recoil. Minimize interruptions (<10 seconds).

Rescue breathing: tilt head, lift chin to open airway. Pinch nose. Give breath over 1 second, watch for chest rise. 30:2 ratio (30 compressions to 2 breaths). If untrained or unable to do rescue breathing — compression-only CPR is still highly effective.

AED: turn on, attach pads (right upper chest below clavicle, left lower side below armpit), follow voice prompts. Stand clear during analysis and shock. Resume CPR immediately after

shock.

Continue CPR until: EMS arrives and takes over, person recovers (starts breathing normally and moves), a second responder or AED takes over, or you are too exhausted to continue safely.

3. Choking

Mild choking: person can cough, breathe, and speak. Encourage forceful coughing. Do not interfere with back blows unless coughing fails.

Severe choking (adult): cannot breathe, cough, or speak. Universal sign — hands clutching throat. Back blows: lean person forward, give 5 firm blows between shoulder blades with heel of hand. Check after each blow.

Abdominal thrusts (Heimlich maneuver): stand behind, make fist just above navel (below ribcage), grasp with other hand. Pull sharply inward and upward 5 times. Alternate with 5 back blows.

Special situations: pregnant/obese — chest thrusts (centre of breastbone) instead of abdominal thrusts. Infant choking (under 1 year): 5 back blows face-down on your forearm, then 5 chest thrusts face-up. Never abdominal thrusts in infants.

If person becomes unconscious: lower to ground, call 112, start CPR (compressions may dislodge object). Each time you open airway before rescue breaths, look for visible object — remove only if clearly visible (never blind finger sweeps).

4. Severe Bleeding

Direct pressure is the first and most important step. Apply firm continuous pressure to the wound with a clean cloth, dressing, or bandage. Maintain pressure for at least 10-15 minutes without lifting to check. If cloth soaks through, add more on top.

Elevation: raise the injured limb above heart level to reduce blood flow to the wound. Continue direct pressure simultaneously.

Wound packing: for deep wounds where direct pressure is difficult (groin, axilla, neck) — pack gauze firmly into wound and apply pressure. Haemostatic gauze (CeloX, QuikClot) is superior

to plain gauze if available.

Tourniquet: for limb bleeding that cannot be controlled with direct pressure. Apply 5-7 cm (2-3 inches) above the wound (NOT over a joint). Tighten until bleeding stops. Note and write the time of application prominently. A tourniquet correctly applied should cause significant pain — a painless tourniquet is too loose. Do not remove in first aid setting.

Assess for shock: pale/ashen skin, cold clammy skin, rapid weak pulse, rapid breathing, confusion, thirst, nausea. Lay patient flat, elevate legs (unless spinal/chest injury), keep warm, give nothing by mouth, call 112. Blood loss of >750 ml causes early shock signs; >1500 ml causes severe shock.

5. Anaphylaxis (Severe Allergic Reaction)

Anaphylaxis is a severe, potentially life-threatening systemic allergic reaction. Common triggers: foods (peanuts, tree nuts, shellfish, milk), medications (penicillin, aspirin, NSAIDs, contrast media), insect stings (bees, wasps), latex.

Signs: skin — urticaria (hives), angioedema (facial/throat swelling), flushing; respiratory — stridor, wheeze, dyspnoea, hoarseness; cardiovascular — hypotension, rapid weak pulse, dizziness, syncope; GI — nausea, vomiting, abdominal cramps.

Epinephrine (adrenaline) auto-injector (EpiPen): inject into outer mid-thigh (can be given through clothing). Standard dose: adult 0.3 mg (EpiPen), children 0.15 mg (EpiPen Jr). Hold for 10 seconds. Repeat after 5-15 minutes if no improvement (second dose if available).

Position: if hypotensive/dizzy — supine with legs elevated. If breathing difficulty — sitting up. If unconscious — recovery position. Never leave standing (cardiovascular collapse risk).

Call 112 immediately even if epinephrine given (effects temporary, biphasic reaction may occur in 4-12 hours). Hospital observation for minimum 4-6 hours after anaphylaxis. Patients should be discharged with prescription for 2 EpiPens and anaphylaxis action plan.

6. Burns

Burn assessment: Rule of Nines (adult) — head 9%, each arm 9%, chest 18%, back 18%, each leg 18%, perineum 1%. Palm (with fingers) = 1% body surface area (BSA). Children: Lund-Browder chart (larger head proportion).

Burn depth: Superficial (1st degree) — redness, pain, intact skin (sunburn). Partial thickness (2nd degree) — blistering, wet/weeping, very painful. Full thickness (3rd degree) — charred/leathery/waxy, painless (nerve destruction). 4th degree — down to bone/tendon.

First aid for burns: STOP the burning process (remove heat source, douse flames, remove burning clothing — not stuck fabric). COOL — cool running water for minimum 20 minutes within 3 hours of injury. Do NOT use ice, iced water, butter, toothpaste, oil, or yoghurt.

COVER — wrap loosely in cling film (single layer), non-stick dressing, or clean plastic bag. Do NOT use fluffy cotton, towels, or adhesive dressings. Do not pop blisters. Remove jewellery near burned area before swelling.

Seek emergency care for: burns >1% BSA in children or >5% BSA in adults, full-thickness burns, burns on face/hands/feet/genitals/major joints, circumferential burns, inhalation injury (singled nasal hair, hoarse voice, carbonaceous sputum). Give paracetamol for pain.

7. Head Injury and Spinal Injury

Minor head injury: brief loss of consciousness (<30 min), post-traumatic amnesia <24 hours, GCS 13-15 post-resuscitation. Observe for: worsening headache, vomiting, confusion, unequal pupils, weakness, CSF from nose/ear (indicates base of skull fracture).

Red flags requiring immediate CT and/or admission: GCS <13, focal neurological deficit, post-traumatic seizure, suspected skull fracture (haematoma over mastoid — Battle's sign, periorbital haematoma — raccoon eyes), penetrating injury, coagulopathy.

Suspected spinal injury: occurs with high-energy trauma (fall from height, MVA, diving injury), elderly low-energy falls. Precautions: do NOT move patient unnecessarily. Inline stabilization of head and neck. Log roll technique for turning. C-spine collar if available.

Unconscious trauma patient: assume spinal injury until proven otherwise. Head-tilt/chin-lift is modified to jaw thrust to open airway while maintaining c-spine neutrality. Primary survey takes priority — an uncontrolled airway kills faster than a missed c-spine injury.

8. Fractures, Dislocations and Sprains

Assessing a fracture: pain, swelling, deformity (angulation, shortening), bruising, inability to use the limb, crepitus (grating sound — do not deliberately elicit), neurovascular deficit (check

pulse, sensation, capillary refill distal to fracture).

First aid for fractures: immobilize the fracture in the position found. Improvised splint using any rigid material (rolled newspaper, wood plank). Pad well. Extend beyond the joints above and below the fracture. Ice pack (wrapped) for swelling. Elevate. Analgesia.

Open (compound) fracture: bone pierces skin. Do NOT attempt to push bone back. Cover with sterile/clean dressing moistened with saline. Maintain immobilization. High infection risk — requires urgent surgical debridement and IV antibiotics. Call 112.

Dislocations: shoulder (most common), patella, finger, hip (high energy). Do NOT attempt reduction in field (risk of vascular/nerve injury). Immobilize in position of comfort.

Neurovascular assessment every 15-30 min. Urgent hospital. Exception: austere environment dislocations by trained providers.

Sprains: PRICE — Protection, Rest, Ice (20 min every 2-3 hours, wrapped), Compression (elastic bandage), Elevation. Grade 3 sprains (complete ligament rupture) need orthopaedic assessment.

9. Medical Emergencies

Diabetic emergencies: Hypoglycemia — if conscious, give 15g glucose (juice, sugar, tablets).

Unconscious — do NOT give by mouth, IV dextrose or IM glucagon, call 112.

Hyperglycemia/DKA — unable to manage at home, hospital required.

Seizures: protect from injury (clear area, cushion head, turn on side). Do NOT restrain, put anything in mouth, or give water/food. Time the seizure. Call 112 if: >5 minutes, second seizure without recovery, known non-epileptic, first seizure, post-ictal confusion prolonged, or occurs in water/pregnancy.

Drowning: call 112. Remove from water if safe to do so. Check breathing. If not breathing — give 5 rescue breaths first (unlike land CPR), then 30:2 CPR. Remove wet clothing, prevent hypothermia. All drowning survivors need hospital evaluation even if apparently recovered.

Hypothermia: remove wet clothing, gently move to warm environment. Warm the core first (warm fluids, warm blankets centrally — armpits, groin, neck). Do NOT massage limbs vigorously (cold peripheral blood to core can drop core temperature further). Hypothermic cardiac arrest — CPR continues; 'not dead until warm and dead' (rewarming can restore circulation in severe hypothermia).

Heat stroke: core temperature $>40^{\circ}\text{C}$ with neurological symptoms (confusion, seizures, loss of consciousness). Emergency cooling: cool water immersion or evaporative cooling (wet sponge + fan). Ice packs to groin, armpits, neck. Target temp $<38.5^{\circ}\text{C}$. Call 112 urgently.

10. Poison, Bites and Stings

General poisoning: identify poison if possible (preserve container/packaging). Do NOT induce vomiting (contraindicated for corrosives, hydrocarbons, or altered consciousness). Call 112 or Poison Control (India: 1800-116-117). Activated charcoal (50g adult, 1g/kg child) within 1-2 hours of ingestion if person alert — in hospital setting.

Snake bite: most deadly medical emergency in India. Stay calm, limit movement of affected limb (splint if possible). DO NOT: cut and suck, apply tourniquet, give traditional remedies, apply ice. Identify the snake if safe to do so (photograph at safe distance). Reach hospital within 1-2 hours. Anti-Snake Venom (ASV) is the definitive treatment.

Scorpion sting: painful with localised symptoms. Red Scorpion (*Mesobuthus tamulus*) in India can cause autonomic storm — hypertension, tachycardia, sweating, pulmonary oedema, myocarditis. Hospital admission mandatory for Red Scorpion stings. Prazosin (alpha-blocker) is the specific treatment in Indian protocol. Scorpion antivenom available in endemic areas.

Dog bite / Rabies prevention: wash wound vigorously with soap and water for 15 minutes. This single measure reduces rabies transmission risk by 90%. Anti-Rabies Vaccine (ARV) — WHO Essen protocol: Days 0, 3, 7, 14, 28 (5 doses). Category III exposure (bite on head/neck/fingers, deep wound, multiple bites) also requires Rabies Immunoglobulin (RIG) on Day 0. Rabies is 100% fatal once symptomatic — prevention is everything.

Jellyfish sting: rinse with sea water (not fresh water — activates more stingers). Remove tentacles with tweezers or a card (do not use bare hands). Immerse in hot water (as hot as tolerable) for 20-45 minutes — inactivates venom. Apply lidocaine gel if available. Seek emergency care for anaphylaxis, cardiovascular symptoms, or box jellyfish (Irukandji) exposure.